

Practical List for the Python for Data Science

B.E CComputer Engineering - 5th Sem

Practical 1: To Load and Explore the Unstructured Access Log Data

Objective: Learn how to import and explore raw server access logs (Apache/Nginx format).

- Load a sample `access.log` file using Python (e.g., with `open()` or `pandas` + regex).
- Understand the typical log structure (IP, timestamp, request, status, size, user-agent, etc.).
- Extract 5–10 random lines to manually inspect the pattern.
- Identify useful fields for analysis.

Practical 2: To Convert the Unstructured Log Data into a Structured Dataset

Objective: Parse log data into structured tabular form.

- Use regex or `apachelogs`, `pyparsing`, or `logparser` libraries to extract fields:
 - IP address
 - Date/time
 - Request type (GET, POST)
 - Status code
 - Resource requested
 - Bytes sent

- Referrer
- User-agent
- Store data in a Pandas DataFrame.

Practical 3: To clean the data and Preprocessing for further process of dataset.

Objective: Clean structured data for analysis.

- Convert timestamp to datetime format.
- Handle missing values and inconsistent entries.
- Normalize URL paths (e.g., `/index.html` vs `/index`).
- Lowercase strings and remove extraneous spaces.

Practical 4: To label the requests as Benign or Attack (Basic Classification) from the dataset.

Objective: Create labeled categories for analysis.

- Use known attack patterns (e.g., SQL injection, path traversal, brute-force login) to label logs.
- Example:
 - SQLi: URLs with `' OR 1=1 --`
 - Brute force: Repeated login attempts from same IP
 - Path traversal: `../` in the URL
- Create a new column: `label` → `benign`, `sqli`, `brute_force`, etc.

Practical 5: To do the feature engineering for the anomaly detection or any suspicious activity.

Objective: Create meaningful features for machine learning.

- Number of requests per IP
- Time between requests
- Status code frequency (e.g., many 404 → suspicious)
- User-agent parsing (bots vs real browsers)
- URL entropy (suspicious randomness)

Practical 6: To do the balancing of the Dataset for ML training/Deep Learning training.

Objective: Handle imbalanced dataset for fair ML training.

- Analyze distribution of benign vs attack traffic.
- Use techniques like:
 - Undersampling of benign
 - Oversampling of attack (SMOTE or random oversampling)
- Prepare a balanced dataset.

Practical 7: To do the Data Wrangling for Aggregated Analysis on the balanced dataset.

Objective: Aggregate and reshape data for deeper insights.

- Group by IP to analyze frequency of attacks
- Time-series resampling (e.g., hourly/daily traffic)
- Use pivot tables to see request type by label
- Filter out bots and internal IPs

Practical 8: To use the visual tools or by using python for the Data Visualization and EDA

Objective: Use visual tools to understand the dataset.

- Plot:
 - Requests per hour
 - Top 10 attacking IPs
 - Attack categories over time
 - Status code distributions
 - Heatmaps of IP vs request types
- Tools: Matplotlib, Seaborn, Plotly

Practical 9: To build a Simple Classifier to detect the attacks.

Objective: Train a basic model to detect attacks.

- Use features from practical 5
- Split into train/test
- Apply Logistic Regression or Random Forest
- Evaluate accuracy, confusion matrix

Practical 10: To create a Reusable Data Pipeline for the log file.

Objective: Wrap the entire log processing into a pipeline.

- Automate:
 - Loading logs
 - Parsing & structuring
 - Labeling
 - Preprocessing
 - Feature engineering
- Save outputs as CSV or parquet